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Java SE 27 & JDK 27

Class ConstantCallSite

java.lang.Object
    java.lang.invoke.CallSite
        java.lang.invoke.ConstantCallSite

```
public non-sealed class ConstantCallSite
extends CallSite
```

A `ConstantCallSite` is a `CallSite` whose target is permanent, and can never be changed. An `invokedynamic` instruction linked to a `ConstantCallSite` is permanently bound to the call site's target.

Since:

1.7

Constructor Summary

Constructors

Modifier	Constructor	Description
	ConstantCallSite(MethodHandle target)	Creates a call site with a permanent target.
protected	**ConstantCallSite**(MethodType targetType, MethodHandle createTarge	Creates a call site with a permanent target, possibly bound to the call site itself.

Method Summary

All Methods | **Instance Methods** |

Modifier and Type	Method	Description
final MethodHandle	**dynamicInvoker**()	Returns this call site's permanent target.
final MethodHandle	**getTarget**()	Returns the target method of the call site, which behaves like a `final` field of the `ConstantCallSite`.
final void	**setTarget**(MethodHandle ignore)	Always throws an `UnsupportedOperationException`.

Methods declared in class CallSite

type

Methods declared in class Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Constructor Details

ConstantCallSite

```
public ConstantCallSite(MethodHandle target)
```

Creates a call site with a permanent target.

Parameters:

`target` - the target to be permanently associated with this call site

Throws:

`NullPointerException` - if the proposed target is null

ConstantCallSite

```
protected ConstantCallSite(MethodType targetType,
                           MethodHandle createTargetHook)
                    throws Throwable
```

Creates a call site with a permanent target, possibly bound to the call site itself.

During construction of the call site, the `createTargetHook` is invoked to produce the actual target, as if by a call of the form `(MethodHandle) createTargetHook.invoke(this)`.

Note that user code cannot perform such an action directly in a subclass constructor, since the target must be fixed before the `ConstantCallSite` constructor returns.

The hook is said to bind the call site to a target method handle, and a typical action would be `someTarget.bindTo(this)`. However, the hook is free to take any action whatever, including ignoring the call site and returning a constant target.

The result returned by the hook must be a method handle of exactly the same type as the call site.

While the hook is being called, the new `ConstantCallSite` object is in a partially constructed state. In this state, a call to `getTarget`, or any other attempt to use the target, will result in an `IllegalStateException`. It is legal at all times to obtain the call site's type using the `type` method.

Parameters:

`targetType` - the type of the method handle to be permanently associated with this call site

`createTargetHook` - a method handle to invoke (on the call site) to produce the call site's target

Throws:

`WrongMethodTypeException` - if the hook cannot be invoked on the required arguments, or if the target returned by the hook is not of the given `targetType`

`NullPointerException` - if the hook returns a null value

`ClassCastException` - if the hook returns something other than a `MethodHandle`

`Throwable` - anything else thrown by the hook function

Method Details

getTarget

`public final MethodHandle getTarget()`

Returns the target method of the call site, which behaves like a `final` field of the `ConstantCallSite`. That is, the target is always the original value passed to the constructor call which created this instance.

Specified by:

`getTarget` in class `CallSite`

Returns:

the immutable linkage state of this call site, a constant method handle

Throws:

`IllegalStateException` - if the `ConstantCallSite` constructor has not completed

See Also:

`ConstantCallSite`,
`VolatileCallSite`,
`CallSite.setTarget(MethodHandle)`,
`getTarget()`,
`MutableCallSite.getTarget()`,

VolatileCallSite.getTarget()

setTarget

```
public final void setTarget(MethodHandle ignore)
```

Always throws an UnsupportedOperationException. This kind of call site cannot change its target.

Specified by:

setTarget in class CallSite

Parameters:

ignore - a new target proposed for the call site, which is ignored

Throws:

UnsupportedOperationException - because this kind of call site cannot change its target

See Also:

CallSite.getTarget(),
setTarget(MethodHandle),
MutableCallSite.setTarget(MethodHandle),
VolatileCallSite.setTarget(MethodHandle)

dynamicInvoker

```
public final MethodHandle dynamicInvoker()
```

Returns this call site's permanent target. Since that target will never change, this is a correct implementation of CallSite.dynamicInvoker.

Specified by:

dynamicInvoker in class CallSite

Returns:

the immutable linkage state of this call site, a constant method handle

Throws:

IllegalStateException - if the ConstantCallSite constructor has not completed

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Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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